第23讲. Gauss公式. Stokes公式.

作业: P190. 1, 2, 10, 11. 15

4, 5, 6, 8, 9.

复习: 设 $\Omega \subset \mathbb{R}^3$ 是由光滑或分片光滑的封闭曲面 Σ 围成有界闭区域.

若 $P(x,y,z)$, $Q(x,y,z)$, $R(x,y,z) \in C^1(\Omega)$. 则

$$\oiint_{\Sigma^+} P\,dy\wedge dz + Q\,dz\wedge dx + R\,dx\wedge dy = \iiint_{\Omega} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dxdydz.$$

例1. 设 $\vec{F} = \frac{1}{\sqrt{(x^2+y^2+z^2)^3}}(x,y,z)$. 求 $I = \oiint_{\Sigma^+} \vec{F}\cdot\vec{n}\,dS$. 其中

Σ^+ 为 Σ 外侧. $\vec{n}$ 为 Σ^+ 单位法向量.

① Σ 及其所围区域 Ω 不包含原点.

② Σ 包围区域 Ω 包含坐标原点.

解: ①.

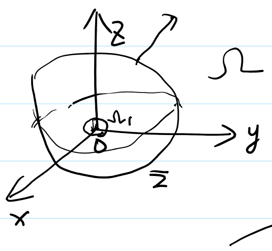
$$P(x,y,z) = \frac{x}{\sqrt{(x^2+y^2+z^2)^3}} \in C^1(\Omega). \qquad \frac{\partial P}{\partial x} = \frac{(x^2+y^2+z^2)^{\frac{3}{2}} - 3x^2(x^2+y^2+z^2)^{\frac{1}{2}}}{(x^2+y^2+z^2)^3} = \frac{y^2+z^2-2x^2}{(x^2+y^2+z^2)^{\frac{5}{2}}}$$

$$Q(x,y,z) = \frac{y}{\sqrt{(x^2+y^2+z^2)^3}} \in C^1(\Omega) \qquad \frac{\partial Q}{\partial y} = \frac{x^2+z^2-2y^2}{(x^2+y^2+z^2)^{5/2}}$$

$$R(x,y,z) = \frac{z}{\sqrt{(x^2+y^2+z^2)^3}} \in C^1(\Omega) \qquad \frac{\partial R}{\partial z} = \frac{x^2+y^2-2z^2}{(x^2+y^2+z^2)^{5/2}}$$

$$\therefore I = \oiint_{\Sigma^+} \vec{F}\cdot\vec{n}\,dS = \iiint_{\Omega} (P_x' + Q_y' + R_z')\,dxdydz = 0.$$

②.



$\Sigma_1 : x^2+y^2+z^2 = \varepsilon^2$.

$\Omega_1 = \{(x,y,z) \mid x^2+y^2+z^2 \le \varepsilon^2\}$. s.t. $\Omega_1 \subset \Omega$. Ω_1 边界为 Σ_1. Σ_1^+ 外侧

$$\because \Omega\setminus\Omega_1 \qquad \oiint_{(\Sigma\cup\Sigma_1)^+} \vec{F}\cdot\vec{n}\,dS = 0.$$

$$\| $$

$$\oiint_{\Sigma^+\cup\Sigma_1^-} \vec{F}\cdot\vec{n}\,dS$$

$$\| $$

$$\iint_{\Sigma^+} \vec{F}\cdot\vec{n}\,dS + \iint_{\Sigma_1^-} \vec{F}\cdot\vec{n}\,dS$$

$$\therefore I = \oiint \vec{F}\cdot\vec{n}\,dS = \oiint \vec{F}\cdot\vec{n}\,dS = \oiint \frac{1}{\sqrt{(x^2+y^2+z^2)^3}}(x,y,z)\cdot\vec{n}\,dS$$

$$\therefore \quad I=\oiint_{\Sigma^+} \vec{F}\cdot\vec{n}\,dS = \oiint_{\Sigma_1^+} \vec{F}\cdot\vec{n}\,dS = \oiint_{\Sigma_1^+} \frac{1}{\sqrt{(x^2+y^2+z^2)^3}}(x,y,z)\cdot\vec{n}\,dS$$

$$=\frac{1}{\varepsilon^3}\oiint_{\Sigma_1^+}(x,y,z)\cdot\vec{n}\,dS = \frac{1}{\varepsilon^3}\cdot\iiint_{\Omega_1} 3\,dxdydz = 4\pi.$$

例2. 计算 $I=\iint_{\Sigma^+} xz^2\,dy\wedge dz + (x^2y-z^2)\,dz\wedge dx + (2xy+y^2z)\,dx\wedge dy$

Σ^+: $z=\sqrt{a^2-x^2-y^2}$. 上侧为正侧.

解: $P(x,y,z) = xz^2 \in C^1$

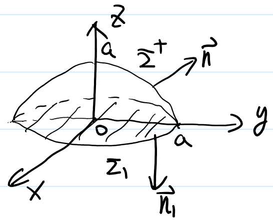
$Q(x,y,z) = x^2y-z^2 \in C^1$

$R(x,y,z) = 2xy+y^2z \in C^1$

$\frac{\partial P}{\partial x} = z^2$

$\frac{\partial Q}{\partial y} = x^2$

$\frac{\partial R}{\partial z} = y^2$



$\vec{n}_1=(0,0,-1)$

Σ_1^+: $\begin{cases} x^2+y^2\le a^2 \\ z=0 \end{cases}$ 向下为正.

$(\Sigma\cup\Sigma_1)^+ = \Sigma^+\cup\Sigma_1^+$

$$I=\iint_{\Sigma^+}\vec{F}\cdot\vec{n}\,dS = \oiint_{(\Sigma\cup\Sigma_1)^+}\vec{F}\cdot\vec{n}\,dS - \iint_{\Sigma_1^+}\vec{F}\cdot\vec{n}_1\,dS$$

$$=\underbrace{\iiint_{\substack{x^2+y^2+z^2\le a^2\\ z\ge 0}}(x^2+y^2+z^2)\,dxdydz}_{I_1} - \underbrace{\iint_{\Sigma_1^+}\vec{F}\cdot\vec{n}_1\,dS}_{I_2}$$

$$\begin{cases} x=\rho\sin\varphi\cos\theta \\ y=\rho\sin\varphi\sin\theta \\ z=\rho\cos\varphi \end{cases} \qquad \rho\in[0,a],\quad \theta\in[0,2\pi],\quad \varphi\in[0,\tfrac{\pi}{2}].$$

$$I_1=\int_0^a d\rho\int_0^{2\pi}d\theta\int_0^{\frac{\pi}{2}}\rho^2\cdot\rho^2\sin\varphi\,d\varphi = \frac{2}{5}\pi a^5.$$

$$I_2=\iint_{\Sigma_1^+} xz^2\,dy\wedge dz + (x^2y-z^2)\,dz\wedge dx + (2xy+y^2z)\,dx\wedge dy$$

$$= \iint_{\Sigma_1^+} (xz^2, x^2y - z^2, 2xy + y^2z) \cdot \vec{n}_1 \, dS$$

$$= -\iint_{\Sigma_1} (2xy + y^2 z)\, dS$$

$$dS = \sqrt{1 + z_x'^2 + z_y'^2}\, dxdy$$

$$= -\iint_{x^2+y^2 \le a^2} 2xy \, dxdy = 0 .$$

$$\therefore \quad I = \frac{2}{5}\pi a^5 .$$

$$\oiint_{\Sigma^+} P\,dy\wedge dz + Q\,dz\wedge dx + R\,dx\wedge dy = \iiint_{\Omega} (P_x' + Q_y' + R_z')\,dxdydz .$$

$P = x. \quad Q = y \quad R = z.$

$$\| $$

$$3\iiint_{\Omega} dxdydz = 3V_{\Omega}$$

$$V_{\Omega} = \frac{1}{3}\oiint_{\Sigma^+} x\,dy\wedge dz + y\,dz\wedge dx + z\,dx\wedge dy$$

例3. 求 $\Omega = \left\{ (x,y,z) \,\middle|\, \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \le 1 \right\}$ 体积.

解: $\Sigma: \begin{cases} x = a\sin\varphi\cos\theta \\ y = b\sin\varphi\sin\theta \\ z = c\cos\varphi \end{cases}$ $\quad \varphi \in [0,\pi]. \quad \theta \in [0, 2\pi].$

$$V_{\Omega} = \frac{1}{3}\oiint_{\Sigma^+} x\,dy\wedge dz + y\,dz\wedge dx + z\,dx\wedge dy$$

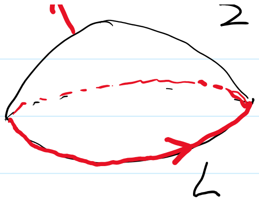
$$= \frac{1}{3}\left| \iint_{\varphi,\theta} (x, y, z)\cdot(A, B, C)\,d\varphi d\theta \right|$$

$$= \frac{1}{3}\left| \iint_{\varphi,\theta} \begin{vmatrix} x & y & z \\ x_\varphi' & y_\varphi' & z_\varphi' \\ x_\theta' & y_\theta' & z_\theta' \end{vmatrix} d\varphi d\theta \right| = \frac{1}{3}\left| \iint_{\substack{\varphi\in[0,\pi] \\ \theta\in[0,2\pi]}} abc\sin\varphi\, d\varphi d\theta \right|$$

$$= \frac{1}{3}\int_0^{2\pi} d\theta \int_0^{\pi} abc\sin\varphi\, d\varphi = \frac{4\pi abc}{3}$$

2. Stokes公式.

右手螺旋法则



右手螺旋法则

定理1. 设 Σ 是分片光滑双侧曲面，其边界是分段光滑封闭曲线 L.

且 $P(x,y,z)$, $Q(x,y,z)$, $R(x,y,z) \in C^1(\Omega)$, Ω 包含 Σ 及 L. 则

$$\oint_{L^+} P\,dx + Q\,dy + R\,dz = \iint_{\Sigma^+} \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right) dy\wedge dz + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right) dz\wedge dx + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\wedge dy$$

$$= \iint_{\Sigma^+} \begin{vmatrix} dy\wedge dz & dz\wedge dx & dx\wedge dy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

其中 Σ^+ 与 L^+ 满足右手螺旋法则.

注：stokes 公式与 Σ 形状无关.

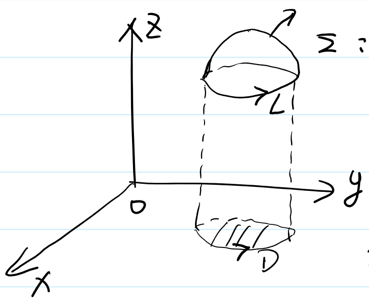
$$\oint_{L^+} P\,dx + \oint_{L^+} Q\,dy + \oint_{L^+} R\,dz$$

$$\oint_{L^+} P\,dx = \iint_{\Sigma^+} \frac{\partial P}{\partial z}\, dz\wedge dx - \frac{\partial P}{\partial y}\, dx\wedge dy$$

$$\oint_{L^+} Q\,dy = \iint_{\Sigma^+} \frac{\partial Q}{\partial x}\, dx\wedge dy - \frac{\partial Q}{\partial z}\, dy\wedge dz$$

$$\oint_{L^+} R\,dz = \iint_{\Sigma^+} \frac{\partial R}{\partial y}\, dy\wedge dz - \frac{\partial R}{\partial x}\, dz\wedge dx$$

证明：1°. Σ 与平行于 z 轴直线交点至多一点.



$\Sigma: z = f(x,y) \in C^1(D)$.

$$\vec{n} = \frac{(-f_x', -f_y', 1)}{\sqrt{\ }}$$

$$\partial D = l: \begin{cases} x = x(t) \\ y = y(t) \end{cases} \quad t: a \to b \quad l^+.$$

$$L^+: \begin{cases} x = x(t) \\ y = y(t) \\ z = f(x(t), y(t)) \end{cases} \quad t: a \to b.$$

$$\oint_{L^+} P\,dx$$

$$\|$$

$$\int_a^b P(x(t), y(t), f(x(t), y(t)))\, x'(t)\, dt$$

$\parallel$

$$\int_a^b P(x(t), y(t), f(x(t), y(t))) x'(t)\, dt$$

$, z = f(x(t), y(t))$

$\parallel$

$$\oint_{l^+} P(x, y, f(x,y))\, dx = -\iint_D (P_y' + P_z' \cdot f_x')\, dx\, dy$$

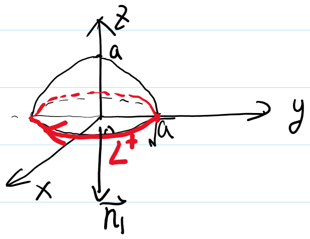
$$\iint_{\Sigma^+} \frac{\partial P}{\partial z} dz\wedge dx - \frac{\partial P}{\partial y} dx \wedge dy = \iint_{\Sigma^+} \left(0, \frac{\partial P}{\partial z}, -\frac{\partial P}{\partial y}\right) \cdot \vec{n}\, dS$$

$$= \iint_D \left[\frac{\partial P}{\partial z} \cdot (-f_y') - \frac{\partial P}{\partial y}\right] dx\, dy$$

例4. 计算 $I = \oint_{L^+} x^2y^3\, dx + dy + dz$.

$L^+ : \begin{cases} x^2+y^2 = a-z \\ z=0 \end{cases}$ 其正向与 z轴成左手螺旋系.

解:



$\Sigma_1^+ : \begin{cases} x^2+y^2 \le a \\ z=0 \end{cases}$

$\vec{n}_1 = (0, 0, -1)$

$$I = \iint_{\Sigma_1^+} \begin{vmatrix} 0 & 0 & -1 \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y^3 & 1 & 1 \end{vmatrix} dS$$

$$= \iint_{\Sigma_1} 3x^2y^2\, dS$$